

Enterprise AI Architect Learning Path v3 – Phase Guide 7

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LLMOps, Observability & Deployment

Purpose of This Phase

The previous phases focused on building increasingly sophisticated AI systems:

- RAG architectures
- Agent orchestration
- MCP tool ecosystems
- Multi-agent workflows

At this point, you have built systems that work.

Phase 7 focuses on making them:

Operable, observable, measurable, maintainable, and deployable.

This is the phase where many AI projects fail.

Organizations often successfully build prototypes but struggle to answer questions such as:

- What is the system doing?
- Why did it fail?
- How much did it cost?
- Which component is slow?
- Which prompt caused the problem?
- Which agent made the decision?
- Is quality improving or degrading?

LLMOps exists to answer those questions.

Why This Phase Matters

Enterprise architects are ultimately responsible for systems operating reliably in production.

A successful AI architecture must provide visibility into:

- System health
- Agent behavior
- Quality trends

- Cost trends
- Infrastructure utilization
- Failure modes

The goal shifts from:

Building AI systems

to:

Operating AI systems responsibly.

Phase Objectives

Upon completion of this phase, you should be able to:

- Implement OpenTelemetry
 - Design AI observability architectures
 - Trace agent workflows
 - Capture AI performance metrics
 - Monitor usage and costs
 - Build evaluation pipelines
 - Design deployment architectures
 - Implement CI/CD workflows
 - Measure system reliability
 - Establish operational support practices
-

Architect Mindset

One of the most important principles of modern AI architecture is:

If you cannot observe it, you cannot govern it.

Enterprise AI systems are inherently probabilistic.

Traditional applications generally fail in predictable ways:

- Exceptions
- Crashes
- Infrastructure outages

AI systems introduce additional failure types:

- Hallucinations

- Retrieval failures
- Tool misuse
- Cost explosions
- Workflow degradation
- Latency spikes

These failures must become visible.

Primary Resources

Required

R16

OpenTelemetry Documentation

Focus Areas:

- Traces
 - Metrics
 - Logs
 - Distributed Observability
-

R17

.NET Observability with OpenTelemetry

Focus Areas:

- Instrumentation patterns
 - Monitoring architecture
 - Telemetry design
-

R3

Azure AI Foundry Documentation

Focus Areas:

- Evaluation
 - Monitoring
 - Application management
-

R21

LangChain Academy

Focus Areas:

- Agent evaluation
 - LangSmith concepts
 - Workflow visibility
-

R2

Designing Machine Learning Systems

Focus Areas:

- Operations
 - Monitoring
 - Production systems
 - Reliability
-

Core Technical Skills

1. Distributed Tracing

A trace tells the story of a request.

Example:

User Question

↓

RAG Retrieval

↓

Search Results

↓

Agent Decision

↓

MCP Tool Call

↓

Final Response

Every major component should contribute telemetry.

2. Metrics

Learn to measure:

System Metrics

Examples:

- Latency
- Throughput
- Error rates

AI Metrics

Examples:

- Prompt counts
- Token usage
- Retrieval scores
- Agent execution counts

Business Metrics

Examples:

- Workflow completions
- Approval rates
- User adoption

3. Logging

Every major event should produce meaningful logs.

Examples:

- Retrieval results
- Agent decisions
- Tool invocations
- Workflow transitions

Logs should support troubleshooting.

4. AI Evaluation Operations

Evaluation should become a continuous process.

Before Deployment

Execute:

- Golden Question suites
 - Regression tests
 - Workflow tests
-

After Deployment

Monitor:

- Quality changes
 - Failure trends
 - User experience
-

5. Cost Management

AI systems introduce new operational concerns.

Track:

Model Cost

Examples:

- Prompt Tokens
 - Completion Tokens
-

Agent Cost

Examples:

- Number of tool calls
 - Workflow length
-

Infrastructure Cost

Examples:

- Search
- Storage
- Compute

Architects must understand both technical and financial behavior.

6. Deployment Architecture

Learn to design systems that can move beyond local development.

Potential targets include:

- Azure App Service
- Azure Container Apps
- Azure Kubernetes Service
- Docker environments

Focus on architectural understanding rather than platform mastery.

AI Observability Architecture

A common observability flow might look like:

Application

↓

OpenTelemetry

↓

Application Insights

↓

Dashboards

↓

Alerting

The objective is visibility rather than complexity.

Phase Project

Instrumented AI Platform

This phase upgrades every previous project.

RPG Catalog

Add:

- Retrieval metrics
 - Search performance
 - Evaluation telemetry
-

Virtual GM

Add:

- Agent traces
 - Tool traces
 - Workflow visibility
-

AI Publishing Company

Add:

- Workflow tracking
 - Agent execution metrics
 - Approval telemetry
-

Functional Requirements

Traceability

Track:

- Requests
 - Decisions
 - Tool calls
 - Workflow states
-

Evaluation Automation

Support:

- Golden Question execution
- Regression testing
- Quality monitoring

Cost Monitoring

Track:

- Usage
 - Cost estimates
 - Consumption patterns
-

Operational Visibility

Enable understanding of:

- What happened
 - When it happened
 - Why it happened
-

Architecture Deliverables

ADR-019

Observability Strategy

Document:

- Telemetry architecture
 - Monitoring approach
 - Visibility requirements
-

ADR-020

Evaluation Program

Document:

- Regression testing
 - Golden question execution
 - Quality measurement
-

ADR-021

Deployment Strategy

Document:

- Hosting options
 - Environment strategy
 - CI/CD approach
-

Portfolio Artifacts

Observability Architecture Diagram

Illustrate:

Applications

↓

Telemetry

↓

Monitoring Platform

↓

Dashboards

Operations Dashboard

Track:

- Usage
 - Latency
 - Errors
 - Quality trends
-

Evaluation Reports

Document:

- Quality metrics
 - Regression results
 - Recommendations
-

Deployment Architecture

Show:

- Hosting model
 - Service interactions
 - Operational boundaries
-

Cost Management Plan

Describe:

- Monitoring strategies
 - Budget controls
 - Optimization approaches
-

Suggested Sprint Progression

Sprint 1 — Observability Planning

Goals:

- Identify telemetry requirements
- Design observability architecture

Done When:

- Strategy documented
-

Sprint 2 — OpenTelemetry Integration

Goals:

- Add tracing
- Add metrics

Done When:

- Telemetry visible
-

Sprint 3 — Logging & Diagnostics

Goals:

- Improve visibility
- Improve troubleshooting

Done When:

- Diagnostics are useful
-

Sprint 4 — Evaluation Automation

Goals:

- Automate testing
- Automate evaluations

Done When:

- Evaluation pipeline operational
-

Sprint 5 — Cost Monitoring

Goals:

- Track usage
- Estimate costs

Done When:

- Cost visibility exists
-

Sprint 6 — Deployment Design

Goals:

- Design hosting architecture
- Create deployment plan

Done When:

- Deployment strategy documented
-

Sprint 7 — Operational Hardening

Goals:

- Improve reliability
- Improve maintainability

Done When:

- Instrumented AI Platform completed
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Common Pitfalls

Pitfall 1

Observability as an Afterthought

Adding telemetry late is much harder than designing it from the beginning.

Pitfall 2

Monitoring Infrastructure Only

AI architectures require visibility into:

- Models
- Retrieval
- Agents
- Workflows

not just servers.

Pitfall 3

No Quality Monitoring

Systems can gradually degrade even while remaining operational.

Quality trends must be measured.

Pitfall 4

Ignoring Costs

AI systems often incur costs in ways traditional systems do not.

Cost visibility is essential.

Pitfall 5

Confusing Logs with Observability

Logs are useful.

Observability includes:

- Logs

- Metrics
- Traces
- Evaluations

All four matter.

Enterprise Translation Layer

The operational patterns developed in this phase apply directly to enterprise environments.

	≡ Portfolio Project	≡ Enterprise Equivalent
1	RPG Catalog Metrics	Knowledge Platform Monitoring
2	Virtual GM Tracing	Agent Workflow Monitoring
3	MCP Tool Telemetry	Service Observability
4	Publishing Workflow Metrics	Business Process Monitoring
5	Golden Question Testing	Production Quality Monitoring
6	Cost Dashboards	AI Financial Governance

The technologies may change, but the operational architecture remains largely the same.

Definition of Done

Phase 7 is complete when:

- OpenTelemetry is implemented
- Traces are visible end-to-end
- Metrics are collected
- Logging is useful and structured
- Evaluation automation exists
- Cost monitoring exists
- Deployment architecture is documented
- Operations dashboards exist
- ADRs are completed
- All portfolio projects are observable and measurable

End-of-Phase Self-Assessment

You are ready for Phase 8 when you can confidently answer:

- Can I trace an AI request from start to finish?
- Can I identify why an AI workflow failed?
- Can I monitor quality over time?
- Can I explain operational costs?
- Can I design deployment architectures?
- Can I operate AI systems rather than simply build them?

If all answers are "yes," proceed to **Phase 8 – Security, Governance & Portfolio Completion**.